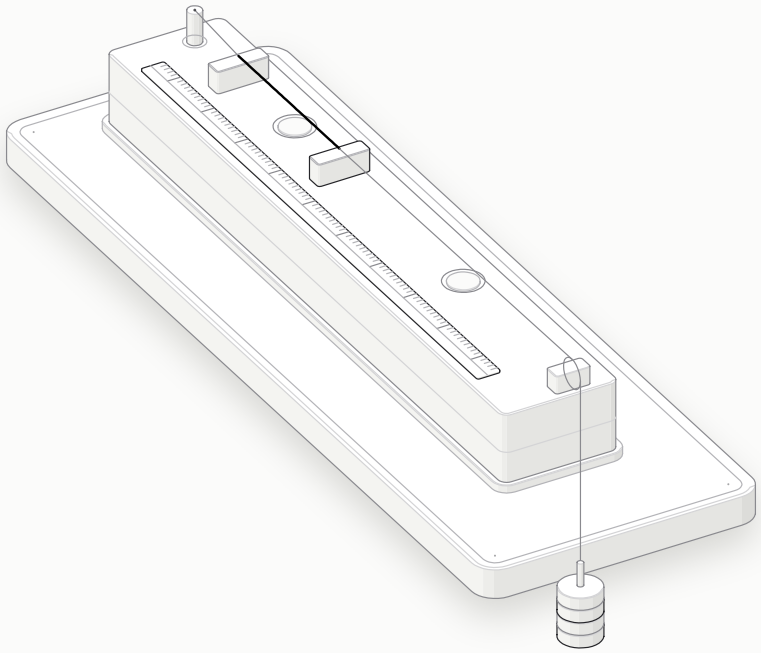
Fig 2

SONOMETER



SLIDE THE BRIDGE

l 30.0 cm · f_1 426 Hz

- Vibrating length: one loop between the bridges, $f_1 = (1/2l)\sqrt{(T/\mu)}$
- Movable bridge: sets l, read on the metre rule

The load hanging over the pulley sets the tension T. Between the bridges the wire's fundamental is $f_1 = (1/2l)\sqrt{(T/\mu)}$. A 256 Hz fork on the box drives the wire; the loop only grows large when the bridge sets l so that f_1 is close to 256 Hz (about 50 cm here), and it is large over a narrow band of lengths, not at a single point.